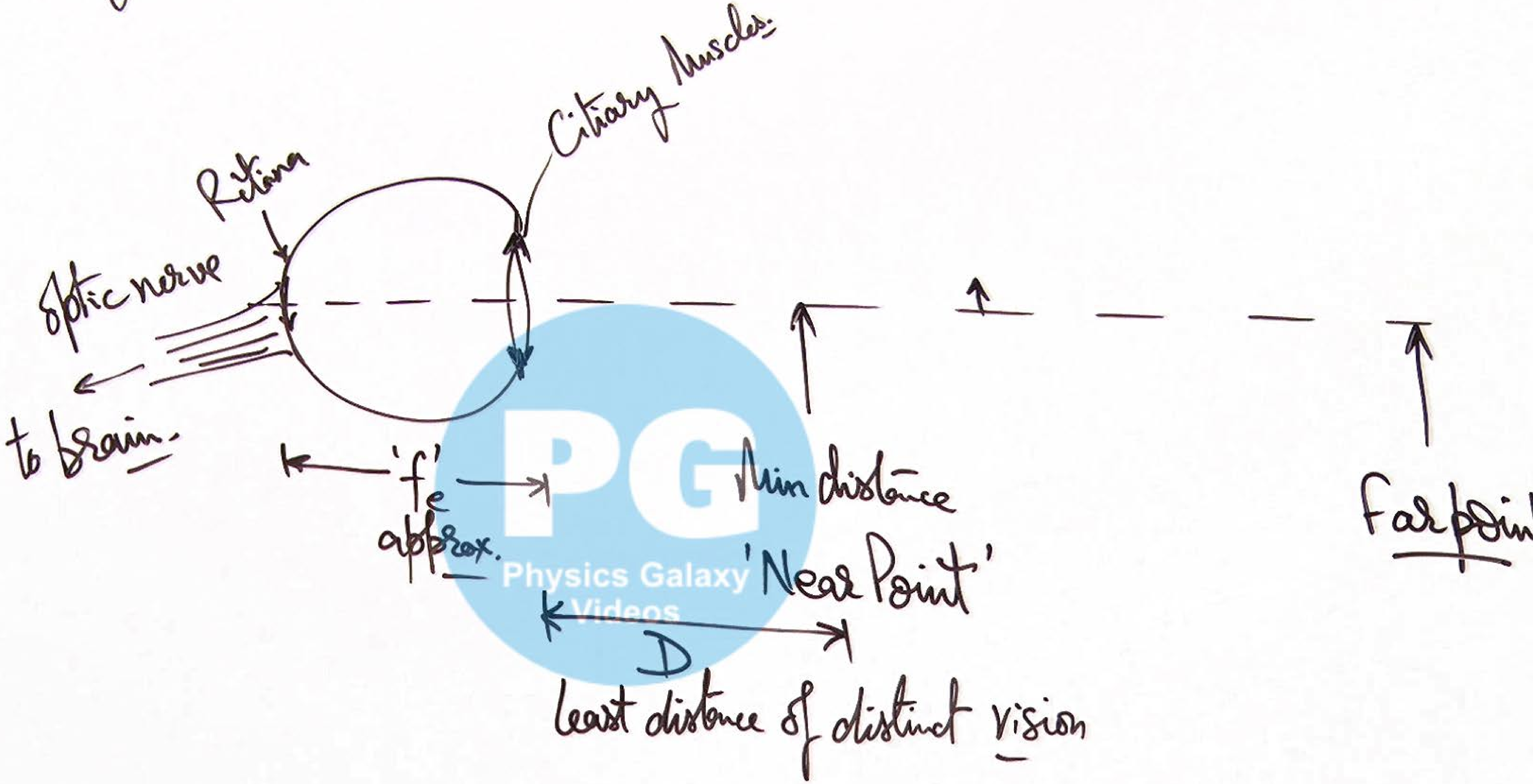
① Types of Optical Instruments :

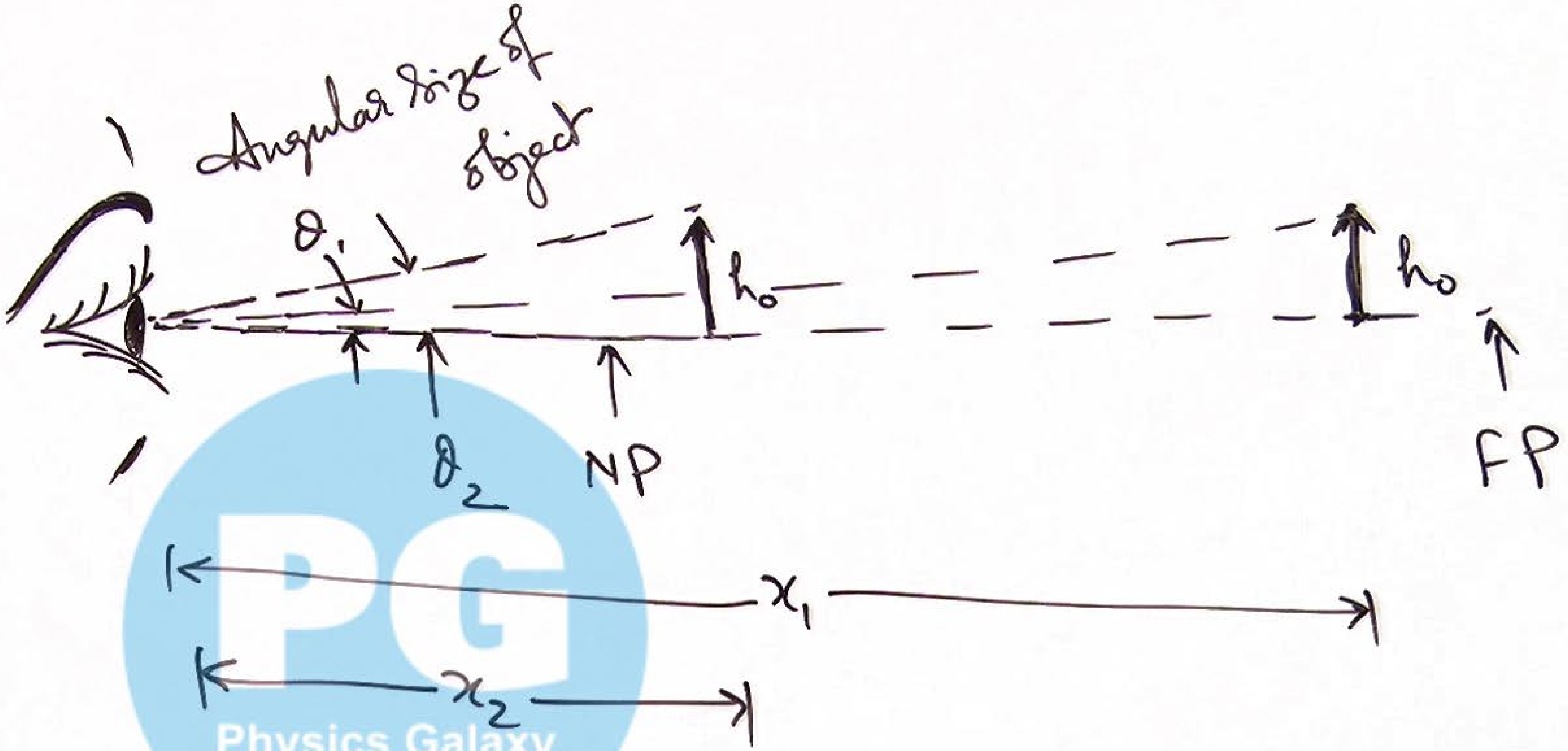
Type-1 : which produce real images of objects

Type-2 : which produce virtual images

(2) Human Eye :


Ciliary Muscles
Retina
Optic nerve
to brain
'f'
f_e
approx.
Min distance
'Near Point'
D
least distance of distinct vision
Far point

③ Angular size of object & Image :



$$\theta_1 = \frac{h_0}{x_1}$$

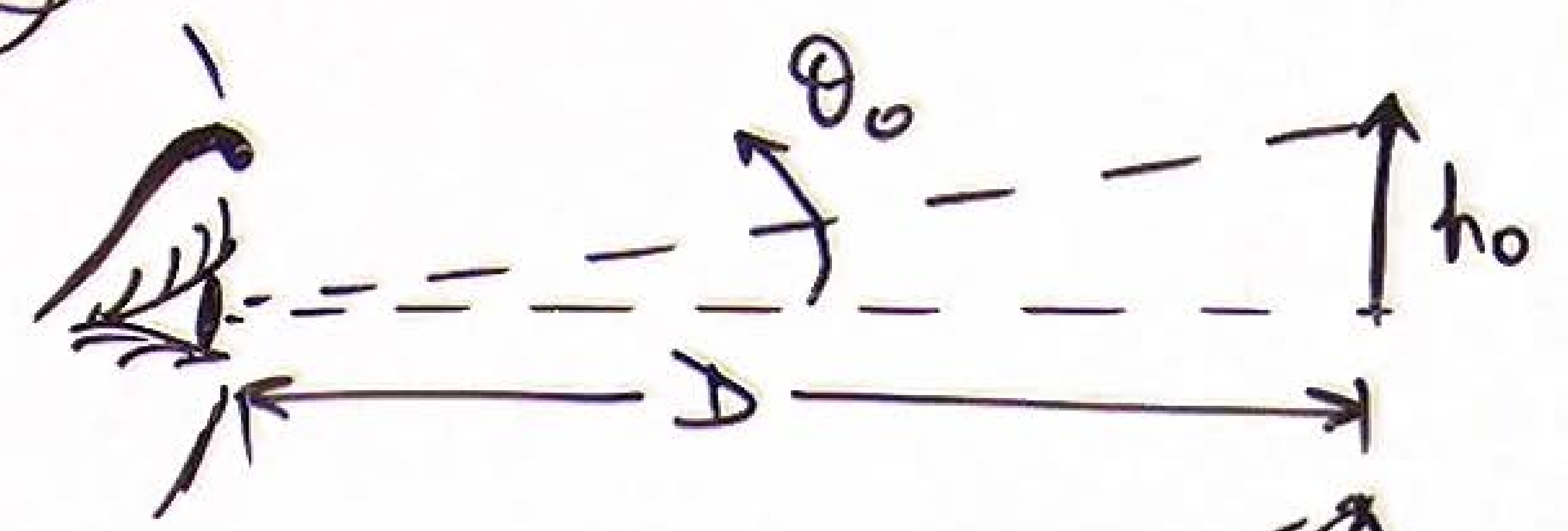
$$\theta_2 = \frac{h_0}{x_2}$$

④ Simple Microscope:

Size of object

$$\theta_0 = \frac{h_0}{D}$$

Base Situation

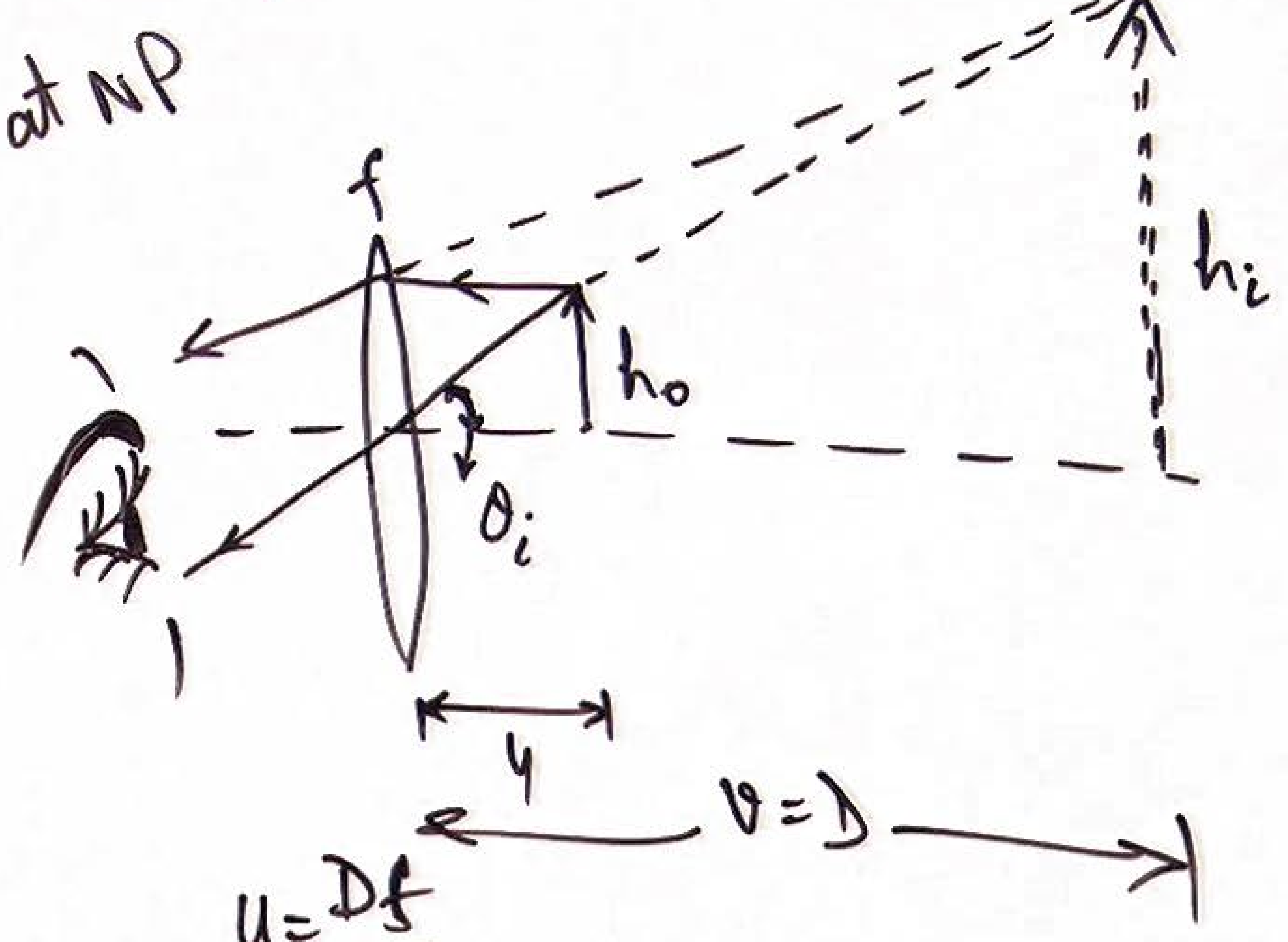


Magnification of a Simple microscope for image at NP

Size of image at NP

$$\theta_i = \frac{h_0}{u}$$

$$\theta_i = \frac{h_0(D+f)}{Df}$$



$$u = \frac{Df}{D+f}$$

for far pt image formation

$$\boxed{m_F = \frac{h_0/f}{h_0/D} = \frac{D}{f}}$$

$$m_N = \frac{\theta_i}{\theta_0} = \frac{h_0(D+f)\cancel{D}}{\cancel{D}f\cdot \cancel{h_0}}$$

$$\boxed{m_N = 1 + \frac{D}{f}}$$

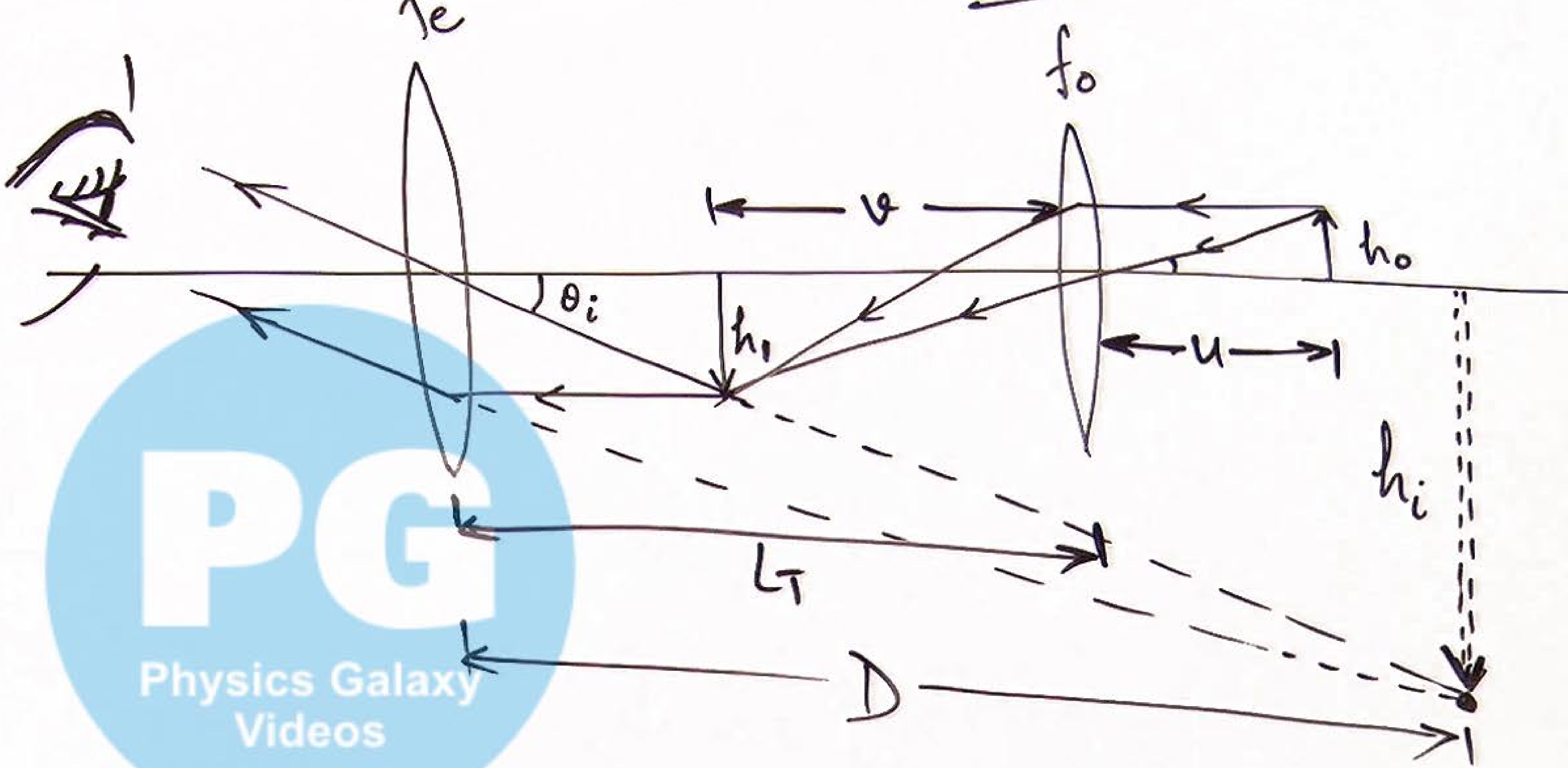
(5) Compound Microscope:

$m_{E_N} = 1 + \frac{D}{f_e}$

$m_{EF} = \frac{D}{f_e}$

$m_o = -\frac{v}{u}$

Magnification of Compound microscope



$m_{CM} = m_E \times m_o$

$$\boxed{m_{N\,CM} = -\frac{v}{u}\left(1 + \frac{D}{f_e}\right)}$$

$$\boxed{m_{F\,CM} = -\frac{v}{u}\left(\frac{D}{f_e}\right)}$$

6) Tube length of Simple Microscope:

$$m_{E_N} = 1 + \frac{D}{f_e}$$

$$m_{EF} = \frac{D}{f_e}$$

$$m_o = -\frac{v}{u}$$

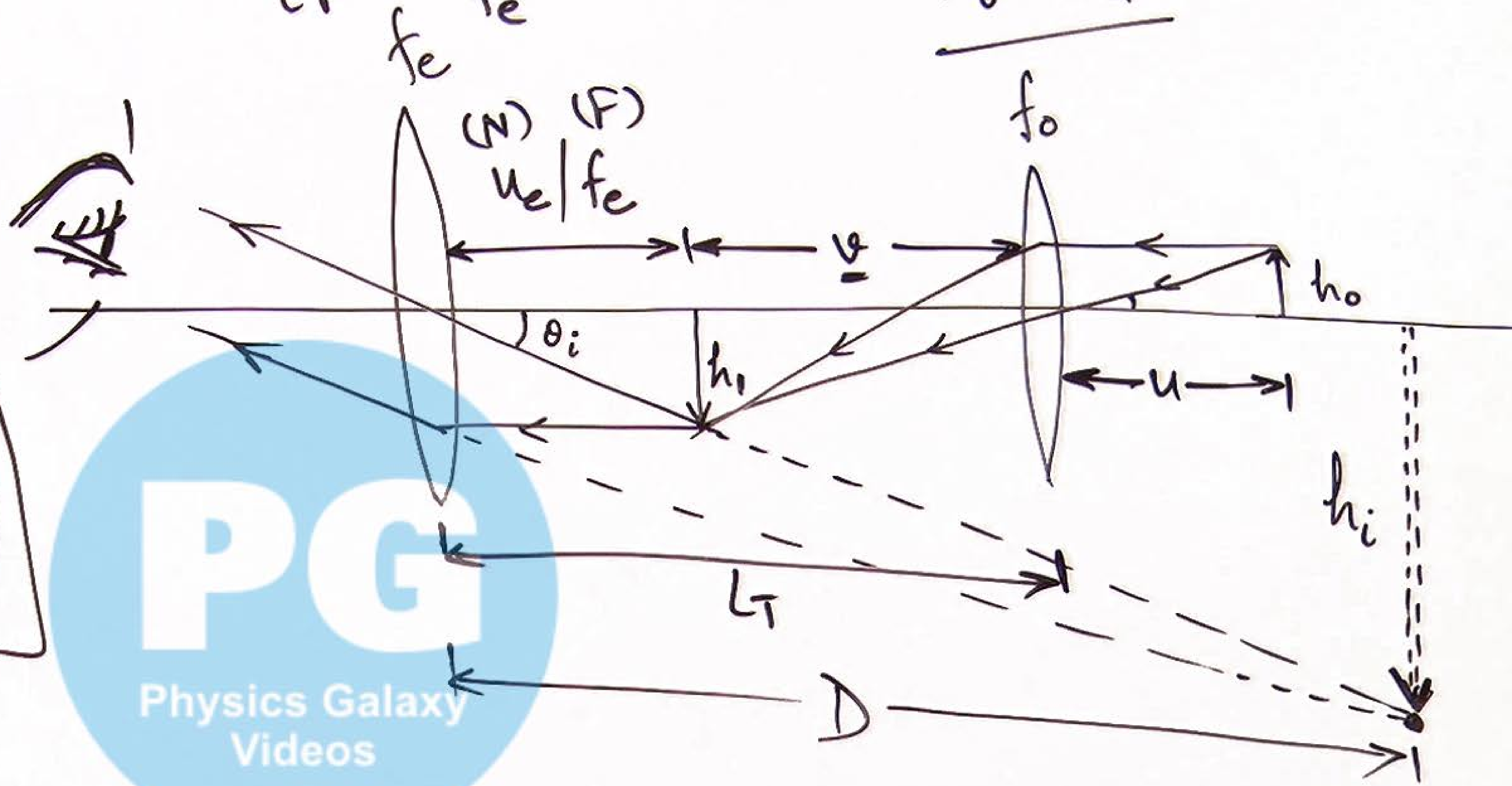
Tube length for NP image production

$$L_{TN} = v + u_e$$

$$\boxed{L_{TN} = \left(\frac{uf_o}{u-f_o}\right) + \left(\frac{Df_e}{D+f_e}\right)}$$

$$L_{TF} = v + f_e$$

$$\boxed{L_{TF} = \frac{uf_o}{u-f_o} + f_e}$$



(8) Refracting Telescope:

Magnification of a R.T. at NP image production

$$m_N = \frac{\theta_i}{\theta_o} = \frac{h_1/u_e}{h_1/f_o}$$

$$m_N = \frac{f_o(D+f_e)}{Df_e}$$

mag of RT at NP img

$$\boxed{m_N = \frac{f_o}{f_e}\left(1+\frac{f_e}{D}\right)}$$

mag of RT at FP img

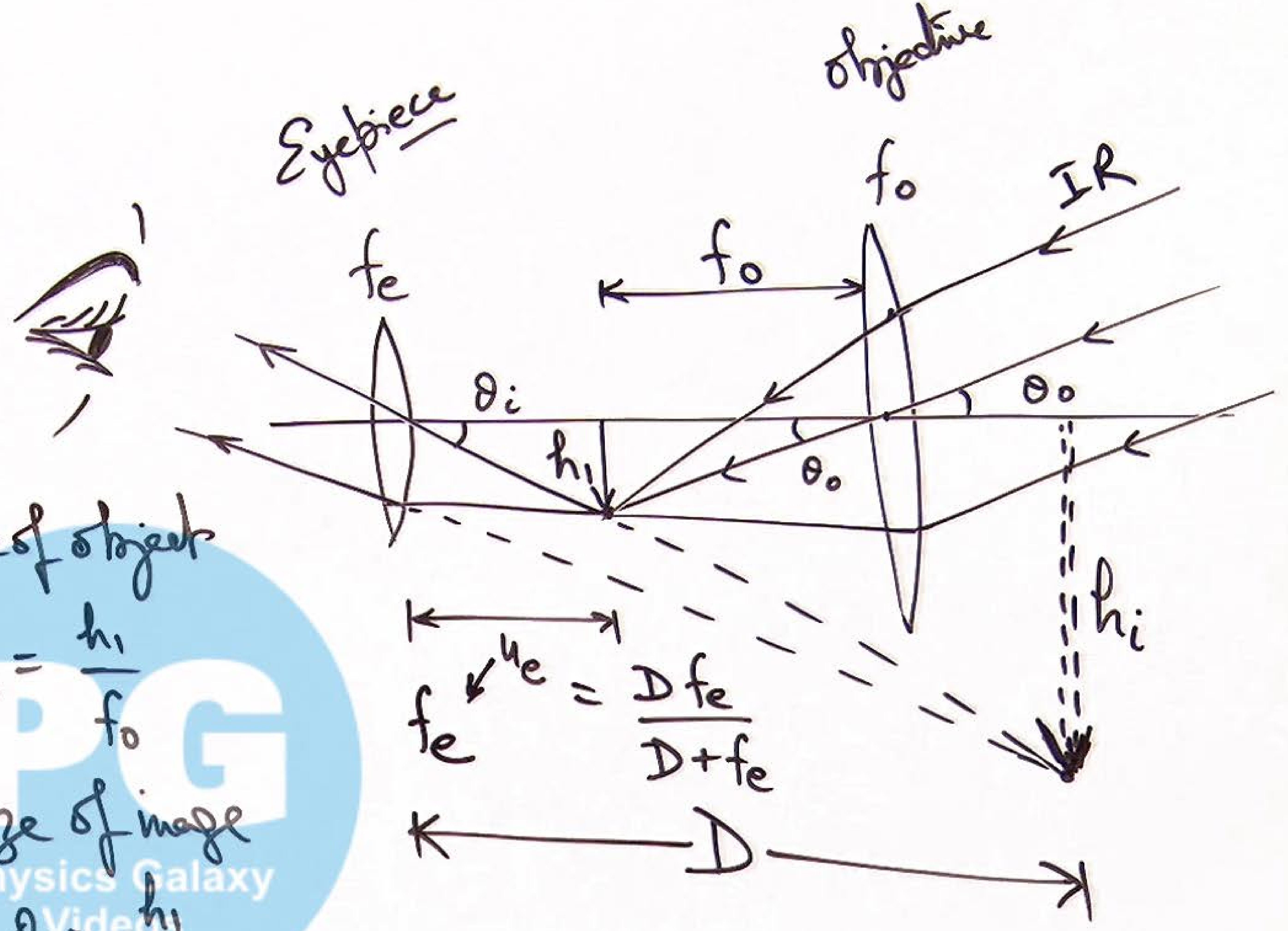
$$\boxed{m_F = \frac{f_o}{f_e}}$$

Ang size of object

$$\theta_o = \frac{h_1}{f_o}$$

Ang size of image

$$\theta_i = \frac{h_1}{u_e}$$



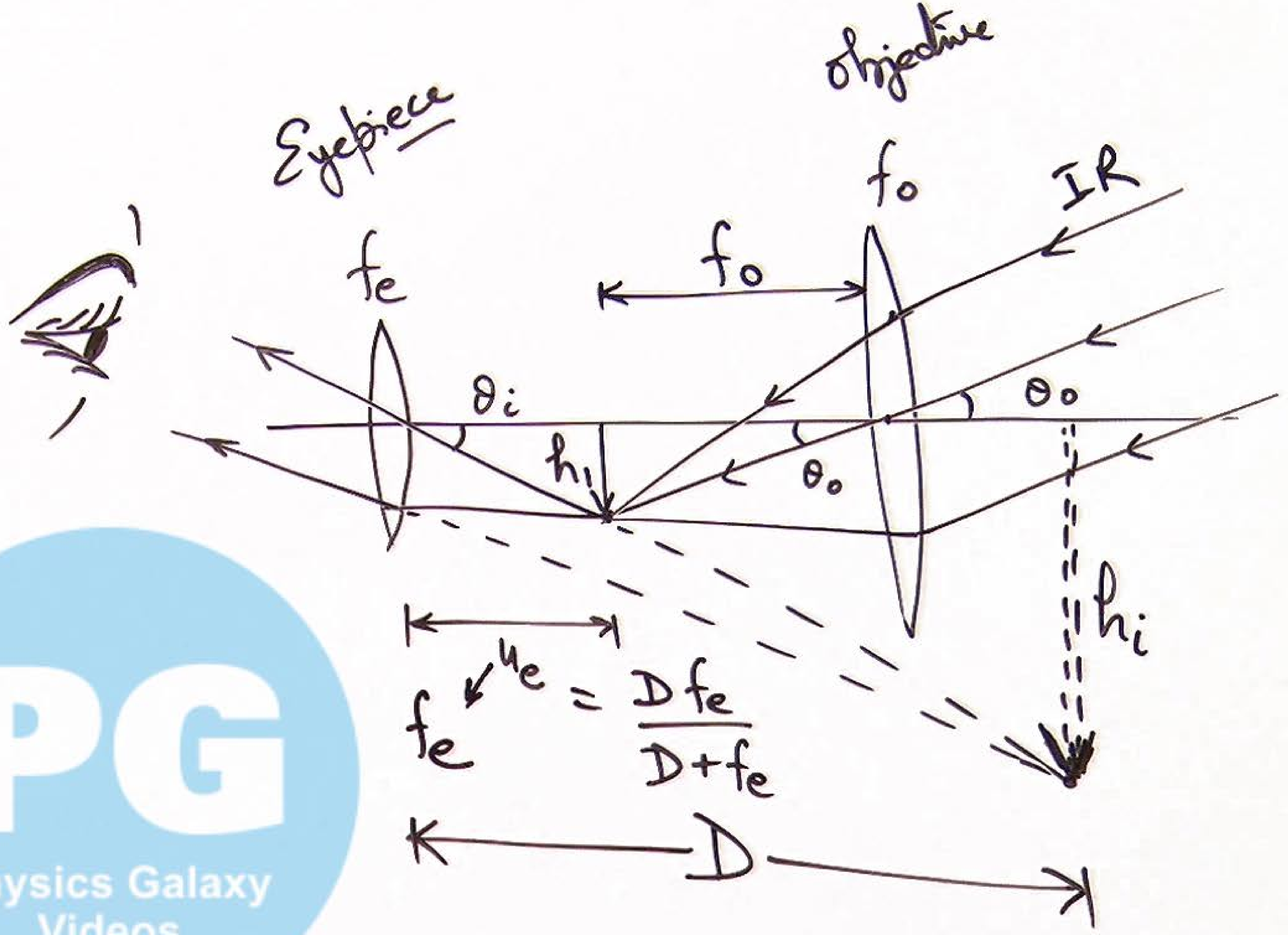
$$u_e = \frac{Df_e}{D+f_e}$$

9 Tube Length of a Refracting Telescope :

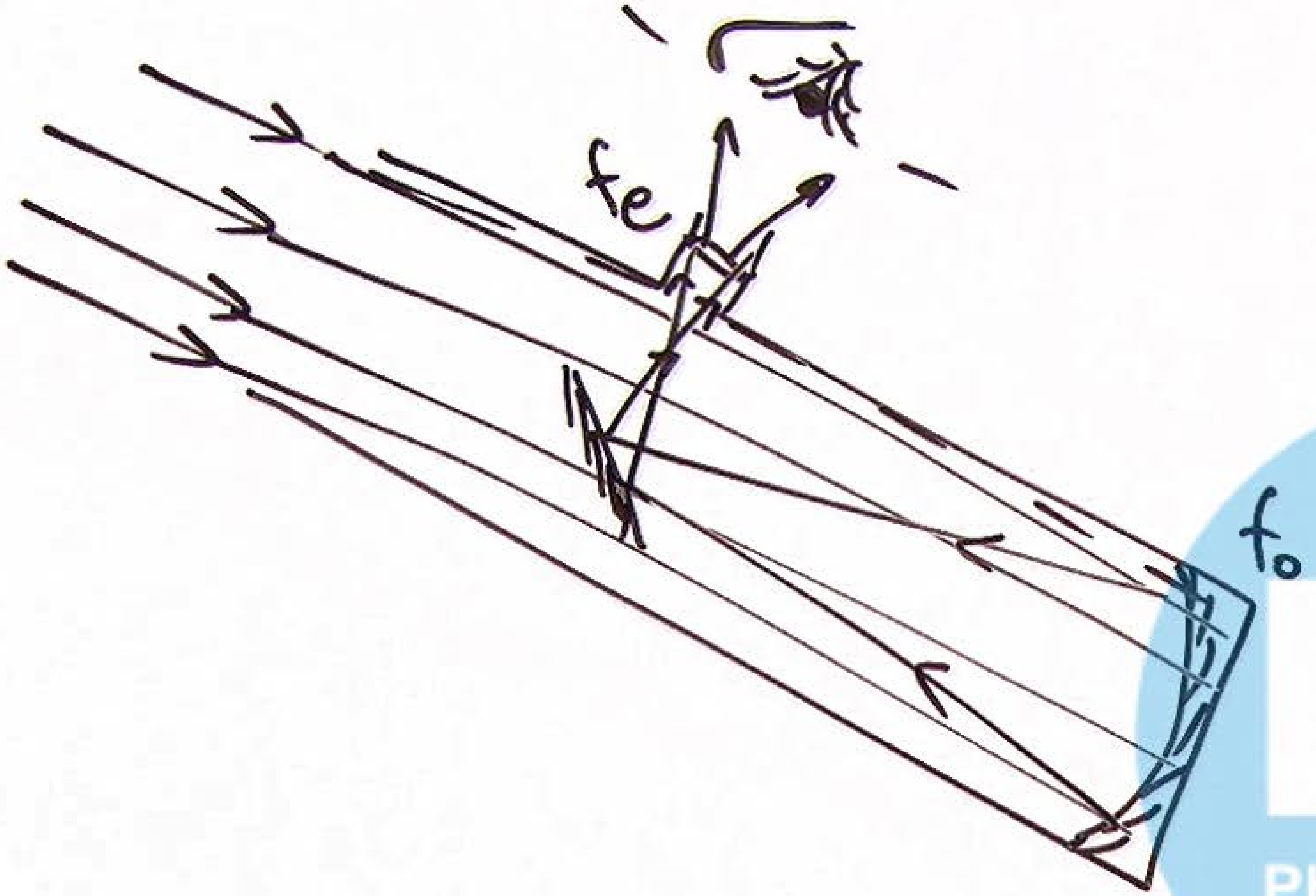
$$L_{TN} = u_e + f_o$$

$$\boxed{L_{TN} = \frac{Df_e}{D+f_e} + f_o}$$

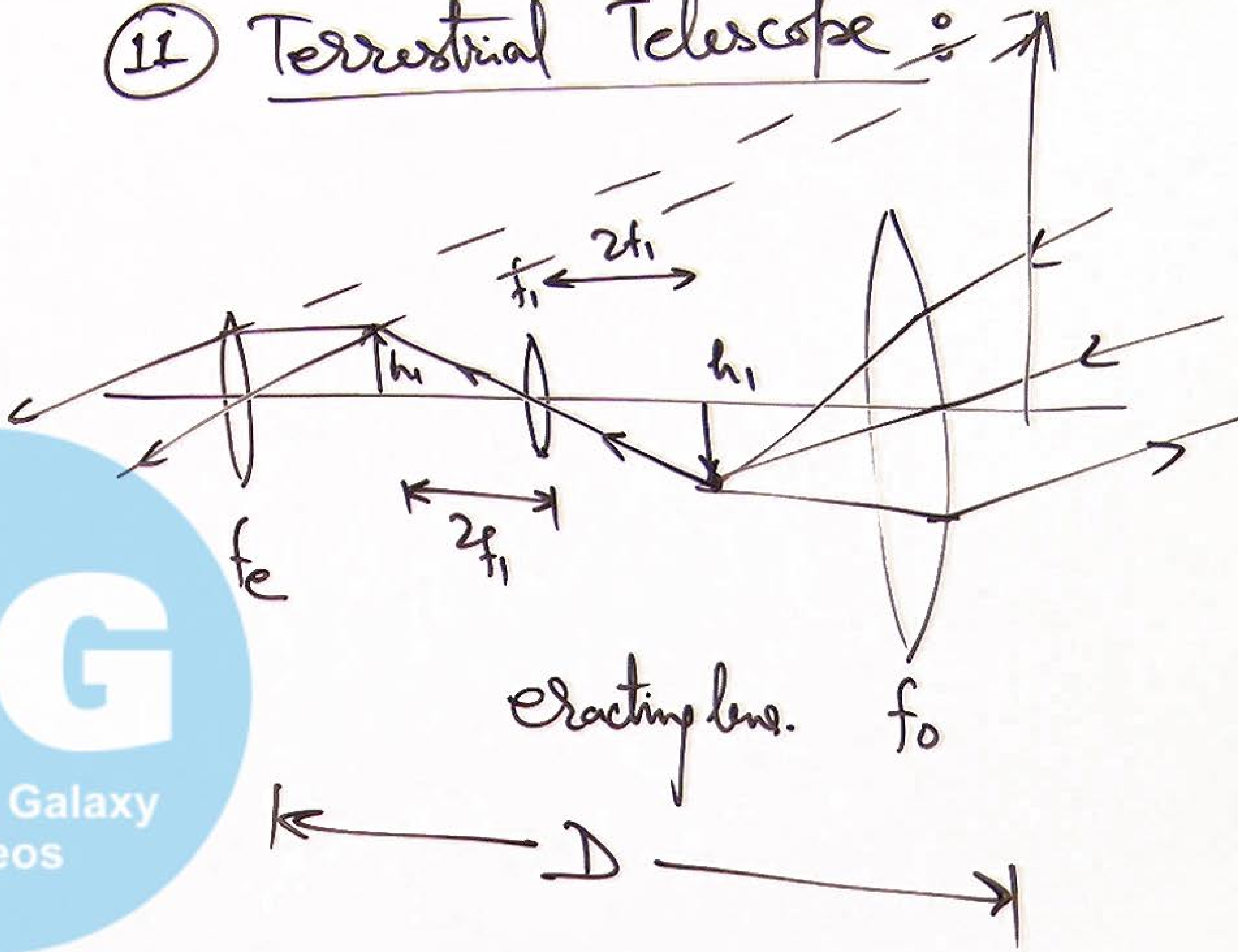
$$\boxed{L_{TF} = f_e + f_o}$$



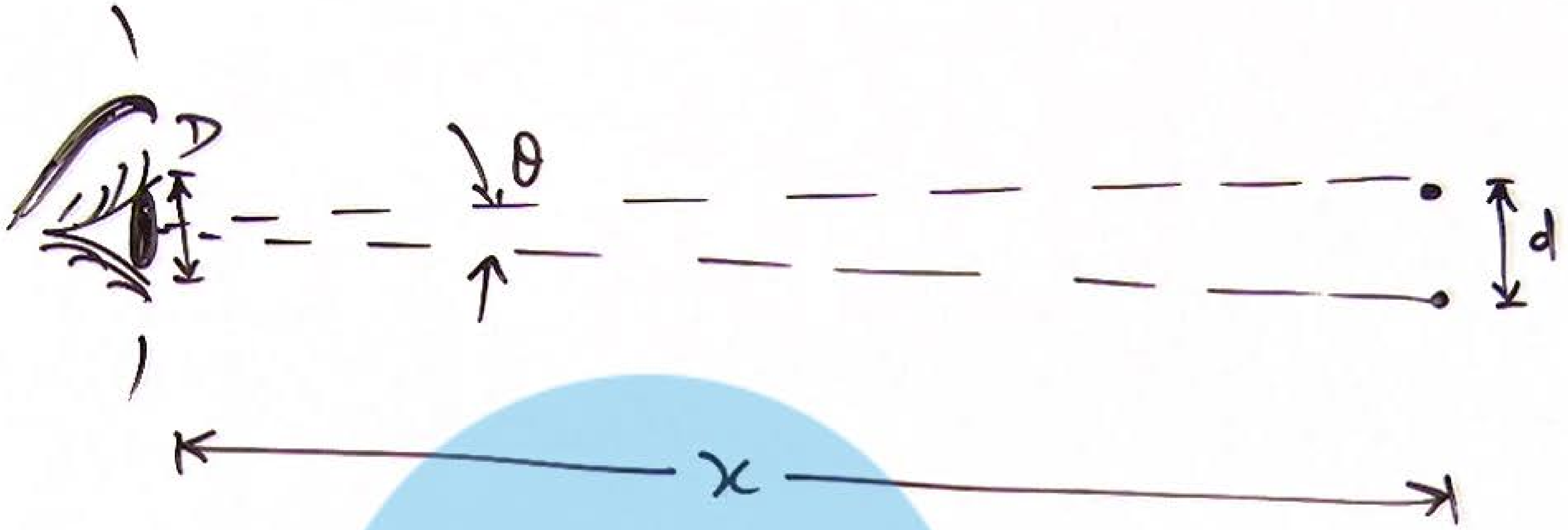
(10) Reflecting Telescope :



(11) Terrestrial Telescope :



11. Limit of Resolution for an Optical Instrument :



$$\theta = \frac{d}{x}$$

Limit of resolution of human eye

$$\boxed{\theta = \frac{1.22\lambda}{D}} = \frac{d}{x} \Rightarrow x = \frac{dD}{1.22\lambda}$$

12. Numerical Aperture (NA)

$$\boxed{NA = \frac{0.61\lambda}{d}}$$

d → Sep between pts to be distinctly seen by OI